Modern Retrieval-Augmented Generation (RAG) Architecture

1. Executive Summary & Overview

Retrieval-Augmented Generation (RAG) has emerged as the industry standard architectural pattern for grounding Large Language Models (LLMs) in external, proprietary enterprise knowledge bases. While LLMs exhibit remarkable reasoning capabilities, they suffer from two critical limitations: parametric knowledge obsolescence and factual hallucinations. In RAG systems, rather than relying solely on static model weights, dynamic external documents are retrieved at inference time and supplied within the LLM context window as verified evidence.

2. The Document Ingestion & Chunking Pipeline

The ingestion pipeline transforms raw unstructured documents (PDFs, Markdown, Docx) into clean, searchable vectors. The standard recommended chunk size for sentence-transformer architectures is between 500 and 1000 characters, paired with a 15% to 20% chunk overlap (typically 100 to 150 characters). Recursive character text splitting is preferred because it respects natural paragraph boundaries, sentence stops, and whitespace hierarchy, thereby preserving local context across contiguous chunk boundaries.

3. Dense Vector Embeddings

Dense embeddings map semantic textual meaning into continuous high-dimensional vector spaces (e.g., 384 dimensions for all-MiniLM-L6-v2, and 1536 dimensions for OpenAI text-embedding-3-small). Unlike sparse lexical methods like BM25 which rely on exact keyword matches, dense vectors capture semantic synonyms, cross-lingual equivalence, and conceptual relationships through cosine similarity.

Vector Databases, Indexing, and Retrieval Dynamics

4. Vector Indexing: FAISS IndexFlatL2 vs. HNSW

Facebook AI Similarity Search (FAISS) provides highly optimized vector operations for GPU and CPU architectures. The IndexFlatL2 index performs exhaustive exact Euclidean distance search with zero recall loss, making it optimal for datasets with fewer than 50,000 vectors. For web-scale vector search exceeding millions of embeddings, Hierarchical Navigable Small World (HNSW) graphs offer logarithmic search complexity O(log N) by traversing multi-layer proximity graphs with approximate nearest neighbor (ANN) guarantees.

5. Retrieval Top-K and Similarity Thresholds

Standard RAG configurations query the vector database for the top-k most similar passages (typically k=3 to k=6). When cosine similarity falls below a critical threshold (e.g., 0.45 or distance > 1.10), the chunks are discarded to prevent context pollution. Context pollution occurs when irrelevant or noisy text confuses the attention mechanism of the generator LLM, leading to degraded synthesis quality.

6. Two-Stage Retrieval with Cross-Encoder Re-Rankers

Production enterprise systems frequently employ a two-stage retrieval pipeline: bi-encoders (such as all-MiniLM-L6-v2) rapidly retrieve the top-50 candidate documents in sub-10ms latency, followed by a computationally heavier cross-encoder (e.g., ms-marco-MiniLM-L-6-v2 or Cohere Rerank) that scores query-document token interactions jointly to select the final top-4 context chunks.

Hallucination Mitigation & Prompt Engineering Protocols

7. Taxonomy of LLM Hallucinations

Hallucinations in generative AI are broadly classified into two categories: Intrinsic hallucinations (where the generated output directly contradicts verified source context) and Extrinsic hallucinations (where the output introduces external claims that cannot be corroborated nor refuted from the source text). In mission-critical legal, medical, and financial domains, extrinsic hallucinations carry severe compliance risks.

8. Prompt Engineering Guardrails for Factual Fidelity

To eliminate ungrounded speculation, production system prompts enforce four fundamental constraints: 1) Absolute domain boundary: The model is explicitly forbidden from utilizing prior parametric memory. 2) Negative rejection protocol: When source evidence is insufficient, the model must output an exact fallback message. 3) Atomic citation requirement: Every declarative statement must conclude with an immutable citation bracket [Source: X, Page: Y]. 4) Low temperature setting: Setting temperature to 0.0 or 0.1 minimizes token sampling stochasticity.

9. Evaluation Frameworks: RAGAS & TruLens

Modern RAG quality assurance evaluates pipelines across four core metrics: Faithfulness (is the answer grounded in context?), Answer Relevance (does the answer address the user query?), Context Precision (are retrieved chunks signal-rich?), and Context Recall (did retrieval capture all necessary background knowledge?). TruLens and RAGAS automate these evaluations.

Enterprise Deployment, Security, and Scalability

10. Latency Optimization and Streaming Inference

User perceived latency is dominated by Time to First Token (TTFT). Utilizing modern inference accelerators such as Groq LPU (Language Processing Unit) architecture enables generation speeds surpassing 500 tokens per second for Llama-3.3-70B models. Streaming response generation allows client interfaces to render partial answers token-by-token, lowering perceived wait times below 300 milliseconds.

11. Document Security & Access Control (RBAC)

When indexing multi-tenant organizational documentation, vector embeddings must retain metadata tags detailing role-based access control (RBAC) permissions. Query filtering is executed at the vector database retrieval stage via metadata pre-filtering, preventing unauthorized documents from ever entering the prompt synthesis context.

12. Conclusion and Future Directions

The evolution of RAG systems is transitioning toward Agentic RAG, where autonomous agents dynamically decide when to retrieve, execute multi-hop reasoning over heterogeneous graph-vector hybrid stores, and self-reflect on output validity before presenting final conclusions to end users.